

Industrial & Systems Engineer (Ph.D.) | Simulation Solution Consultant | AI  
Solution Architect

# JEHUN LEE

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## Professional Profile

In the AI era, the most important question is not which model to use, but what problem is worth solving — defined through human will and necessity. Algorithms don't choose direction; people do.

My direction is to be a keyman: someone who sees the entire system, maps stakeholders, sets priorities, and deeply understands the problems that need solving so direction can be defined. The growth that lasts is the growth where the organization grows together — not where individuals optimize alone.

Systemic excellence, to me, means hiring AI agents — treating them as employees who execute autonomously, with humans setting direction and confirming key decisions. Chores, basic reviews, missing numbers move into agent pipelines, freeing human attention for the work only humans can do: judgment, strategy, meaning.

## Education

|                   |                                                      |                                                            |
|-------------------|------------------------------------------------------|------------------------------------------------------------|
| 2021.03 - 2025.02 | <b>Ph.D. in Industrial &amp; Systems Engineering</b> | Korea Advanced Institute of Science and Technology (KAIST) |
| 2016.09 - 2021.02 | <b>M.S. in Industrial Engineering</b>                | Sungkyunkwan University (SKKU)                             |
| 2013.03 - 2016.08 | <b>B.S. in Systems Management Engineering</b>        | Sungkyunkwan University (SKKU)                             |
| 2011.03 - 2013.02 | <b>High School in Math &amp; Science Track</b>       | Sejong Science High School                                 |

## Work Experience

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|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025.08 - Present | <b>Planning &amp; Scheduling Solution Consultant</b><br>VMS Solutions Inc.   Solution Division   Consultant Part<br>Ministry of National Defense   Tech. Research Personnel [2025.08 - 2026.02]<br>System designer & PMO: digital twin platform, production-logistics integrated simulator, simulation-based AI scheduler, SaaS product<br>– SK Hynix Digital Twin Platform: production-logistics simulator PM, use-case discovery, KPI definition, data consistency metrics, MES legacy data analysis & simulator integration, AI prediction model data specification<br>– Production-logistics integrated simulation design<br>– Production simulation engine architecture redesign & development participation<br>– Planned and authored 2 government R&D proposals on Agentic AI (manufacturing-specific Agentic AI, LLM-based production plan explanation module / with University of Toronto)<br>– AI extensibility review & design for simulation-based APS, new engine feature definition, participation in GenAI-based development<br>– LLM-based schedule analysis & report generation system performance review and feedback |
| 2025.02 - 2025.07 | <b>Planning &amp; Scheduling Solution Consultant</b><br>VMS Solutions Inc.   Solution Division   Micron Part<br>Ministry of National Defense   Tech. Research Personnel [2025.02 - 2025.07]<br>Solution consultant: simulation-based real-time APS maintenance, system configuration remote support<br>– Micron global fab simulation-based real-time APS (advanced production and scheduling) system logic maintenance and remote system configuration support                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

- 2021.03 - 2025.02 **Researcher**  
**KAIST | MSS (Manufacturing Service Systems) Lab.**  
 Ministry of National Defense | Tech. Research Personnel [2023.03 - 2025.02]  
 Lead researcher: AI-based dynamic schedulers, optimization-based heuristics, autonomous manufacturing scheduling systems
- Led full project lifecycle (planning-to-validation): 9 industry-academia projects (planned 5, PM 4), 4 government projects (planned 3, PM 4)
  - Fundamental research on Neural Combinatorial Optimization based on GNN and RL
  - Designed general-purpose scheduling agent and Meta-RL methodology for flexible adaptation to manufacturing environment changes
  - Designed AI-based demand forecasting, production planning, and job scheduling optimization algorithms; validated business impact using real production data (Samsung Electronics, LG Electronics, Samsung Heavy Industries, etc.)
  - Provided technical review supporting client executives' strategic decisions and delivered system design proposals
  - Introduced structured project management processes (OneDrive & Notion-based) to the lab for the first time
  - First Ph.D. graduate of the lab; led project planning, proposal writing, and research beyond typical researcher scope
  - Won 10-year NRF 'Hanwoomul' long-term project after analyzing rejection feedback and reinforcing proposal logic
  - 2 first-author IEEE journal papers, 7 international conference presentations
- 2016.04 - 2021.02 **Researcher**  
**SKKU | SCO (Systems Control & Optimization) Lab.**  
 Scheduling algorithm developer: meta-heuristics, optimization; System designer: digital twin for manufacturing, dynamic rescheduling systems
- 5 industry-academia projects (planned 2, PM 2), 4 government projects (planned 2, PM 1)
  - Led foundational design and operations optimization research for 3D printer-based smart factory digital twin systems
  - Developed rescheduling algorithms for handling disruptions (machine failures, order cancellations) in flexible manufacturing systems (FMS)
  - Translated physical manufacturing constraints into mathematical models and validated in virtual environments
  - Defined virtual factory operation scenarios and conducted simulation-based operational efficiency reviews
  - Designed and verified real-time scheduling methodologies using genetic algorithms and heuristics

## Projects

- 2025.06 - 2025.12 **Digital twin platform construction: production-logistics simulation-based operation twin for semiconductor fab**  
 7m
- Discovered real operation use-cases for the process twin through stakeholder interviews
  - Planned and designed production-logistics integrated simulator: unified modeling of process flows and material movement
  - Integrated real semiconductor fab operational data and performed process twin consistency validation (MES data analysis, SQL)
  - Defined simulation result consistency metrics and designed/developed automated computation pipeline
  - WEB UI design participation and feedback
  - Use-case discovery, KPI definition, consistency metrics definition, automated validation pipeline
- VMS Solutions Inc. (with SK Hynix, SKT, SK AX, Calro)
- 2025.03 - **Software product design/development: simulation-based scheduling for semiconductor manufacturing**  
 - User scenario definition and data schema design  
 VMS Solutions Inc.
- 2025.02 - 2025.06 **Simulation-based real-time scheduler for semiconductor fab**  
 5m (5y)
- Logic design and review for new real-time scheduler: equipment state changes, priority recalculation, and other core algorithm specification
  - Abstracted semiconductor process-specific constraints (machine eligibility, batching rules) for improved product generality
  - Remote solution environment configuration support
  - Resolved Mozart environment issues; designed logic based on customer requirements
- VMS Solutions Inc. (with Micron)
- 2024.05 - 2025.02 **Autonomous scheduling for swift and efficient adaptation to dynamic manufacturing environments**  
 10m (10y)
- Led 10-year project planning and execution
  - Defined progressive extension cases with process constraints and multi-objective functions
  - Year 1: Developed GNN-based imitation learning methodology for multi-process mixed-flow production that adapts to varying problem sizes
  - Makespan optimal gap improved by 42.0% (vs. previous SOTA AI); 1 SCI journal paper (1st author)
- KAIST (with NRF of Korea)
- 2024.03 - 2025.02 **AI-based algorithm for optimal operation plans for various scenarios**  
 1y
- Identified decision-making processes through on-site engineer interviews
  - Defined problem scope and resolution boundaries
  - Data preprocessing and calibration
  - Forecasting model conceptualization, discussion, and refinement
  - Data segmentation
  - Demand forecasting model accuracy: 89.3% / 80% / 80%
- KAIST (with Samsung Electronics)

- 2023.07 - 2024.06  
1y
- Reinforcement learning for unrelated parallel machine scheduling problems with sequence-dependent setup times and machine eligibility**
- Formalized complex manufacturing constraints (eligibility, sequence-dependent setups, batching) into an RL-tractable form
  - Designed a GNN state representation encoding both machine-operation eligibility and setup-time dependencies
  - Trained and evaluated the RL agent across diverse instances; built the benchmark framework vs classical heuristics
  - Consolidated findings into deliverables for VMS Solutions' semiconductor APS roadmap
  - Delivered a GNN-RL methodology that natively handles machine eligibility and sequence-dependent setups — the first such direct treatment in this problem class. Results fed into VMS Solutions' simulation-based semiconductor scheduling product roadmap.
- KAIST (with VMS Solutions)
- 2023.07 - 2024.02  
8m
- Production planning with AI**
- Identified on-site constraints and defined problem through engineer interviews (objective: minimize setup count, maximize on-time production and equipment utilization)
  - Data analysis and calibration (Excel → Python)
  - Developed optimal production planning algorithm (optimization engine-based heuristic)
  - Defined algorithm-based production planning process
  - 7-day plan with 20min runtime limit: setup reduced 5.1%, on-time production improved 25.8%, idle time reduced 37.6% (vs. on-site manual plan)
- KAIST (with LG Electronics)
- 2022.05 - 2023.02  
10m
- Graph-based reinforcement learning algorithm for real-time job shop scheduling**
- Built a high-fidelity simulator for Job Shop environments
  - Designed the GNN-based state representation
  - Trained and evaluated the imitation-learning-based RL dispatcher
  - Discussed applicability to other manufacturing environments (battery, automotive, shipbuilding)
  - Established the GNN-based imitation-learning dispatcher methodology. Foundation for the IEEE Transactions on Automation Science and Engineering first-author paper (2024).
- KAIST (with NRF of Korea)
- 2022.04 - 2023.03  
1y
- Reinforcement learning for job shop scheduling**
- Enhanced the Job Shop simulator to reflect diverse constraints
  - Designed and implemented an AI-based dispatching methodology
  - Reviewed the applicability of imitation learning
  - Fed findings back into the VMS Solutions product roadmap
  - Established an AI-based Job Shop dispatching methodology. Resulted in the WSC 2022 first-author paper "Imitation Learning for Real-Time Job Shop Scheduling using Graph-based Representation".
- KAIST (with VMS Solutions)
- 2022.03 - 2022.03  
1m (7m)
- Development of a wiring optimization algorithm for X-DEC slim layout**
- Identified X-DEC Slim Layout wiring constraints through on-site engineer interviews
  - Defined the problem and resolution scope
  - Reviewed combinatorial-optimization-based methodologies and delivered technical recommendations
  - Accumulated experience analyzing semiconductor memory-cell layout optimization through a combinatorial-optimization lens.
- KAIST (with SK Hynix)
- 2022.03 - 2025.02  
3y
- Reinforcement learning-based meta-scheduling for manufacturing systems**
- Designed a Meta-RL framework on graph-based state representations that generalizes across varying problem sizes and operation sequences
  - Built a tree-based look-ahead search (SBBS) combining beam search with branch-and-bound for tight decision-time budgets
  - Adopted an active-schedule-based RL training scheme to cut wasted search over dominated solutions
  - Benchmarked against fixed-environment RL baselines and classical dispatching rules under environment-change scenarios
  - SBBS was proved mathematically to always produce schedules equal to or better than dispatching-only solutions in static environments. The methodology generalized across problem sizes without retraining and formed the theoretical core of the doctoral thesis, awarded 2nd Prize at the KIIE Ph.D. Thesis Competition (2025.11).
- KAIST (with NRF of Korea)
- 2022.03 - 2023.07  
1y5m (2y)
- Reinforcement learning for project scheduling**
- Led project planning and proposal writing
  - Defined problem and resolution scope (objective: minimize daily worker employment deviation)
  - Enhanced ship production simulator
  - Enhanced RL-based schedule adjustment algorithm
  - Proposed iterative greedy algorithm
  - Discussed spatial layout algorithm methodology
  - Year 1 schedule adjustment: daily manpower std. dev. reduced by 41.60% (RL, 20.27s), 44.53% (greedy, 49.86s), 47.69% (greedy+RL) vs. existing 6-month schedule
- KAIST (with Samsung Heavy Industries)

- 2021.07 - 2021.11  
5m
- Development of a reinforcement learning algorithm for workload balancing of ship cargo production**
- Modeled cargo-hold production constraints through shipyard engineer interviews
  - Designed a project-based ship-production simulator reflecting real yard operations
  - Compared three algorithms — pure RL, pure Greedy, and a Greedy+RL hybrid
  - Benchmarked performance against the baseline 6-month schedule and analyzed convergence speed
  - Daily manpower standard deviation dropped vs the baseline schedule — RL 41.60% (20.27 s), Greedy 44.53% (49.86 s), Greedy+RL 47.69%. RL converged ~2.4× faster than iterative heuristics. Won Samsung Heavy Industries Certificate of Appreciation (2022.05) and 2nd Prize at the KAIST IE Poster Competition (2022.09). KAIST (with Samsung Heavy Industries)
- 2020.06 - 2021.02  
9m
- Optimal machine assignment with machine learning algorithms**
- Analyzed setup-change patterns in semiconductor equipment
  - Designed and trained an ML-based setup-change prediction model
  - Built a machine-assignment algorithm that consumes the prediction output
  - Validated performance via simulation-based benchmarking
  - Implemented an AI-prediction-driven optimal machine-assignment algorithm. Foundation of the WSC 2021 first-author paper "Machine Learning-based Periodic Setup Changes for Semiconductor Manufacturing Machines". SKKU (with VMS Solutions)
- 2019.06 - 2022.05  
3y
- Cyber-physical assembly and logistics systems in global supply chains**
- Identified on-site constraints through field interviews
  - Defined problem and resolution scope
  - Developed automotive parts production simulator and real-time worker allocation/reallocation algorithm
  - Developed conveyor belt speed determination algorithm for semi-automated assembly processes
  - Defined user scenarios and program screen specifications
  - Cycle time reduced 5.6% (real on-site data, actual operation method) / 2-5% (vs. meta-heuristic algorithm); Patent registered: KR1020210070359; 1 SCI journal paper (2nd author); 2 conference papers KAIST (with MOTIE of Korea, Yura)
- 2019.09 - 2020.02  
6m
- Optimal weight sets for dispatching rules with multiple KPIs**
- Defined the multi-KPI hierarchy and priority structure
  - Extended the Sequential Search framework to multi-objective settings
  - Ran simulations across diverse weight scenarios and analyzed results
  - Consolidated outputs for VMS Solutions product integration
  - Completed the multi-objective Sequential Search framework. Resulted in the WSC 2020 first-author paper "A Simulation-based Sequential Search Method for Multi-objective Scheduling Problems" — culminating the dispatching-rule research line with the IEEE TSM 2020 first-author publication. SKKU (with VMS Solutions)
- 2019.04 - 2019.12  
9m (1y9m)
- Big data-based simulation and optimization technology for smart manufacturing**
- Identified battery production constraints through on-site engineer interviews
  - Built a battery production simulator and defined the resolution scope
  - Designed scheduling algorithms reflecting process-specific characteristics
  - Defined MES-integration data schema for the production operations system
  - Established a simulation-based framework that supports real-time decision-making under large, noisy production data. Built collaborative experience with Micube Solution, Samsung SDI, and KITECH. SKKU (with MOTIE of Korea, Samsung SDI)
- 2019.03 - 2022.02  
3y
- Development of scheduling theory and algorithms with reinforcement learning for manufacturing systems**
- Built a high-fidelity simulator for Job Shop environments
  - Designed and implemented an RL-based scheduling methodology
  - Discussed applicability to other manufacturing environments (battery, automotive, shipbuilding)
  - Wrote academic papers and presented at domestic and international conferences
  - Secured foundational know-how on RL-based scheduling. Foundation for the WSC 2022 first-author paper "Imitation Learning for Real-Time Job Shop Scheduling using Graph-based Representation". SKKU (with NRF of Korea)
- 2018.07 - 2018.09  
3m
- Methodology for dispatching rules' weights**
- Identified FAB-line constraints through on-site engineer interviews
  - Analyzed correlations among multiple dispatching rules
  - Applied the Sequential Search framework to the SK Hynix FAB environment
  - Consolidated validation results and identified follow-up research directions
  - Demonstrated that the Sequential Search methodology — developed in the VMS Solutions × SKKU research thread — applies to real semiconductor FAB environments, culminating in the IEEE TSM (2020) first-author publication. SKKU (with SK Hynix)
- 2018.07 - 2019.02  
8m
- Framework development for KPI analysis with various weights on dispatching rules**
- Designed a KPI-change analysis framework keyed on dispatching-rule weight perturbations
  - Built an ML-based algorithm for deriving optimal weight configurations
  - Built a simulation-based validation environment for repeatable benchmarking
  - Documented results for VMS Solutions product integration
  - Established a framework that quantitatively evaluates and derives dispatching-rule weights. Foundation for the WSC 2019 first-author paper. SKKU (with VMS Solutions)

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2017.07 -<br>2018.02<br>8m        | <b>Analysis of KPIs according to weights on dispatching rules for LCD manufacturing</b> <ul style="list-style-type: none"> <li>– Analyzed LCD process characteristics and organized the dispatching-rule matrix</li> <li>– Built a simulation-based performance-comparison environment</li> <li>– Analyzed and visualized weight-KPI correlations</li> <li>– Established a performance-analysis framework keyed on dispatching-rule weight changes. Foundation for the WSC 2018 first-author paper "A Framework for Performance Analysis of Dispatching Rules". SKKU (with VMS Solutions)</li> </ul>                                                                                                                                                     |
| 2017.07 -<br>2018.01<br>7m        | <b>Design and analysis for operations optimizations of smart factory testbed</b> <ul style="list-style-type: none"> <li>– Defined virtual-factory operation scenarios and built the simulator</li> <li>– Conducted simulation-based operational efficiency analysis</li> <li>– Consolidated findings and identified follow-up research directions</li> <li>– Established a methodology for validating operational efficiency through simulation. SKKU (with MSIP of Korea)</li> </ul>                                                                                                                                                                                                                                                                    |
| 2016.11 -<br>2019.10<br>3y        | <b>Development of scheduling and rescheduling algorithms for 3D printer-based smart factory</b> <ul style="list-style-type: none"> <li>– Analyzed characteristics of 3D-printer-based flexible manufacturing systems</li> <li>– Defined disruption scenarios — machine failure, order cancellation, rush orders</li> <li>– Developed rescheduling algorithms for disruption response</li> <li>– Designed real-time schedule derivation using genetic algorithms and heuristics, and validated performance</li> <li>– Built a rescheduling mechanism that responds in real time to machine state and order changes. Resulted in a 2nd-author publication in Int. Journal of Advanced Manufacturing Technology (2020). SKKU (with NRF of Korea)</li> </ul> |
| 2016.07 -<br>2017.02<br>8m        | <b>Development of algorithms for detecting and improving inefficient schedules in LCD processes</b> <ul style="list-style-type: none"> <li>– Identified inefficient schedule patterns through LCD operational data analysis</li> <li>– Defined inefficiency-detection metrics and automated their validation</li> <li>– Designed and implemented heuristic improvement algorithms</li> <li>– First industrial research engagement with VMS Solutions — established the starting point of the multi-year dispatching-rule research line. SKKU (with VMS Solutions)</li> </ul>                                                                                                                                                                             |
| 2016.04 -<br>2018.05<br>2y2m (3y) | <b>Development of open FaaS IoT service platform for mass personalization</b> <ul style="list-style-type: none"> <li>– Built the 3D-printer Urban Factory scheduling simulator and the initial scheduling algorithm</li> <li>– Developed rescheduling algorithms for disruption response</li> <li>– Reviewed robot-handler operation algorithms</li> <li>– Defined data schema and supported integration of the scheduling algorithm into the product</li> <li>– Accumulated experience operating a large multi-institution government project that integrated distributed heterogeneous equipment under FaaS principles. SKKU (with MSIP of Korea)</li> </ul>                                                                                           |

## Academic Works

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|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 2024 | <b>Graph-based imitation learning for real-time job shop dispatcher</b><br>IEEE Transactions on Automation Science and Engineering                          | 1st Author<br><a href="#">[LINK]</a> |
| 2024 | <b>Tree-based Dispatcher for Job Shop Scheduling</b><br>IEEE International Conference on Automation Science and Engineering (CASE)                          | 1st Author                           |
| 2024 | <b>Tree-based dispatcher for solving job shop scheduling problems</b><br>the Spring Conference of Korean Institute of Industrial Engineers                  | 1st Author                           |
| 2023 | <b>Active schedule-based imitation learning for job shop scheduling</b><br>the Spring Conference of Korean Institute of Industrial Engineers                | 1st Author                           |
| 2022 | <b>Imitation learning for real-time job shop scheduling using graph-based representation</b><br>Winter Simulation Conference                                | 1st Author<br><a href="#">[LINK]</a> |
| 2022 | <b>Job shop scheduling using graph-based imitation learning</b><br>INFORMS Annual Meeting                                                                   | 1st Author                           |
| 2022 | <b>A multi-manned assembly line worker assignment and balancing problem with positional constraints</b><br>IEEE Robotics and Automation Letters             | Co-Author<br><a href="#">[LINK]</a>  |
| 2022 | <b>Reinforcement learning for resource leveling in multiple projects</b><br>the Spring Conference of Korean Institute of Industrial Engineers               | 1st Author                           |
| 2022 | <b>Dynamic job shop scheduling using graph-based imitation learning</b><br>the Spring Conference of Korean Institute of Industrial Engineers                | 1st Author                           |
| 2021 | <b>Machine learning-based periodic setup changes for semiconductor manufacturing machines</b><br>Winter Simulation Conference                               | 1st Author<br><a href="#">[LINK]</a> |
| 2021 | <b>Resource leveling in shipyard cargo hold process through reinforcement learning</b><br>the Autumn Conference of Korean Institute of Industrial Engineers | Co-Author                            |

|      |                                                                                                                                                                                  |                                      |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 2021 | <b>Assembly line worker assignment and balancing problem with positional constraints</b><br>Advances in Production Management Systems (APMS)                                     | Co-Author<br><a href="#">[LINK]</a>  |
| 2021 | <b>Operation and optimization of the automotive parts assembly line considering worker skill levels</b><br>the Summer Conference of Korea CDE                                    | Co-Author                            |
| 2020 | <b>A sequential search method of dispatching rules for scheduling of LCD manufacturing systems</b><br>IEEE Transactions on Semiconductor Manufacturing                           | 1st Author<br><a href="#">[LINK]</a> |
| 2020 | <b>A simulation-based sequential search method for multi-objective scheduling problems of manufacturing systems</b><br>Winter Simulation Conference                              | 1st Author                           |
| 2020 | <b>Workforce assignment for automotive parts assembly lines</b><br>the Winter Conference of Korea CDE                                                                            | Co-Author                            |
| 2020 | <b>Digital twin-based cyber physical production system architectural framework for personalized production</b><br>The International Journal of Advanced Manufacturing Technology | Co-Author<br><a href="#">[LINK]</a>  |
| 2020 | <b>Workforce assignment with a different skill level for automotive parts assembly lines</b><br>Advances in Production Management Systems (APMS)                                 | Co-Author<br><a href="#">[LINK]</a>  |
| 2019 | <b>A sequential search framework for selecting weights of dispatching rules in manufacturing systems</b><br>Winter Simulation Conference                                         | 1st Author<br><a href="#">[LINK]</a> |
| 2019 | <b>A genetic algorithm for hybrid flow shop scheduling with multiple assembly stages</b><br>the Autumn Conference of Korean Institute of Industrial Engineers                    | 1st Author                           |
| 2018 | <b>Vulnerability analysis of evacuation transportation networks</b><br>International Journal of Industrial Engineering-Theory Applications and Practice                          | Co-Author                            |
| 2018 | <b>A framework for performance analysis of dispatching rules in manufacturing systems</b><br>Winter Simulation Conference                                                        | 1st Author                           |
| 2018 | <b>Rescheduling algorithms for 3D printer-based manufacturing systems</b><br>the Summer Conference of Korea CDE                                                                  | 1st Author                           |
| 2018 | <b>Scheduling algorithms for 3D printer-based manufacturing systems</b><br>the Spring Conference of Korean Institute of Industrial Engineers                                     | Co-Author                            |
| 2017 | <b>Rescheduling of flexible flow shop with sequence-dependent setup times and job splitting</b><br>Winter Simulation Conference                                                  | Co-Author<br><a href="#">[LINK]</a>  |
| 2017 | <b>3D printer based assembly process scheduling algorithm development</b><br>the Winter Conference of Korea CDE                                                                  | Co-Author                            |

## Honors & Awards

|                   |                                                                                                                   |                                                 |
|-------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 2025.11           | <b>Second Prize, Ph.D. Thesis Competition</b> , Learning Schedulers for Job Shop Scheduling Problems              | Korean Institute of Industrial Engineers (KIIE) |
| 2023.12           | <b>First Prize, 2023 Simulation Challenge</b> , Q-time-based Gating Control in Semiconductor Fabrication          | Winter Simulation Conference (WSC), Micron      |
| 2023.09           | <b>Third Prize, 2023 AI Competition: Solving Real-world Problems</b> , Integrated Planning Module using AI        | Hankook & Company                               |
| 2022.06 - 2023.02 | <b>Ph.D. Candidate Research Incentive Support</b>                                                                 | National Research Foundation (NRF) of Korea     |
| 2022.09           | <b>Second Prize, 2022 Poster Competition: Industry/Social Problems</b> , RL for Resource Leveling in Shipbuilding | KAIST                                           |
| 2022.05           | <b>Certificate of Appreciation: Successful Project</b> , Workload Balancing of Ship Cargo Production              | Samsung Heavy Industries                        |
| 2021.03 - 2025.02 | <b>Full-tuition Entrance Scholarship</b> , Merit-based Entrance Scholarship (Ph.D.)                               | KAIST                                           |
| 2019.03, 2019.09  | <b>Graduate Assistant Tuition Waiver</b> , Tuition Waiver (M.S. & Ph.D Integrated Program)                        | SKKU                                            |

|                   |                                                                                                             |                     |
|-------------------|-------------------------------------------------------------------------------------------------------------|---------------------|
| 2018.03, 2019.03  | <b>Simsan Scholarship</b> , Scholarship (M.S. & Ph.D Integrated Program)                                    | SKKU                |
| 2018.03           | <b>Systems &amp; Management Engineering Scholarship</b> , Scholarship (M.S. & Ph.D Integrated Program)      | SKKU                |
| 2017.03           | <b>Industrial Engineering Alumni Association Scholarship</b> , Scholarship (M.S. & Ph.D Integrated Program) | SKKU                |
| 2016.07           | <b>Third Prize, 2015-2016 PACE RSMS Competition (Second Year): Customer Insight</b>                         | General Motors (GM) |
| 2016.07           | <b>Third Prize, 2015-2016 PACE RSMS Competition (Second Year): Manufacturing Engineering</b>                | General Motors (GM) |
| 2013.03 - 2016.08 | <b>Full-tuition Entrance Scholarship</b> , Merit-based Entrance Scholarship (B.S.)                          | SKKU                |

## Activities & Leadership

|                   |                                  |                                                                            |
|-------------------|----------------------------------|----------------------------------------------------------------------------|
| 2022.03 - 2023.08 | <b>Researcher Representative</b> | Manufacturing and Service Systems Lab. (KAIST)                             |
| 2021.10 – 2021.12 | <b>Technical Mentor</b>          | Public Data Internship Program (Korea National Information Society Agency) |
| 2021.03 – 2024.12 | <b>Teaching Assistant (TA)</b>   | Scheduling (KAIST)                                                         |
| 2018.09 – 2019.06 | <b>Teaching Assistant (TA)</b>   | Supply Chain Management (SKKU)                                             |
| 2018.03 – 2019.02 | <b>President</b>                 | Industrial Engineering Graduate Student Council (SKKU)                     |
| 2018.03 – 2018.06 | <b>Teaching Assistant (TA)</b>   | Operations Management (SKKU)                                               |
| 2016.10 – 2018.12 | <b>Teaching Assistant (TA)</b>   | Gender Awareness Education (SKKU)                                          |
| 2017.09 – 2017.12 | <b>Teaching Assistant (TA)</b>   | Engineering Economy (SKKU)                                                 |
| 2017.03 – 2017.06 | <b>Teaching Assistant (TA)</b>   | Operations Research and Practice 1 (SKKU)                                  |
| 2016.09 – 2016.12 | <b>Teaching Assistant (TA)</b>   | Engineering Economy (SKKU)                                                 |
| 2015.03 – 2016.08 | <b>Working Group Member</b>      | Systems Management Engineering Student Council (SKKU)                      |
| 2015.03 – 2016.08 | <b>President</b>                 | Turbo: Basketball Club (SKKU)                                              |
| 2014.08 – 2020.02 | <b>Working Group Member</b>      | Turbo: Basketball Club (SKKU)                                              |
| 2013.03 – 2015.02 | <b>Working Group Member</b>      | College of Engineering Student Council (SKKU)                              |
| 2012.03 - 2013.02 | <b>President</b>                 | Student Council (High School, 11th Grade)                                  |
| 2010.03 - 2011.02 | <b>Vice President</b>            | Student Council (Middle School, 9th Grade)                                 |

## Skills

|                        |                                                            |
|------------------------|------------------------------------------------------------|
| Programming            | Python, C#, SQL, LaTeX, R                                  |
| Frameworks & Libraries | PyTorch, Matplotlib, Pandas, NumPy, Gurobi, CPLEX          |
| Simulation & Tools     | VMS Mozart (APS), Plant Sim, AutoMod, MS Office, Photoshop |
| DevOps & Productivity  | Git / GitHub, Docker, Claude Code, n8n, Notion             |
| Languages              | Korean (Native), English (Professional)                    |